CSL 7640: Natural Language Understanding

Assignment-2 Report

<u>Problem 2: LEARNING WORD EMBEDDINGS FROM IIT JODHPUR DATA</u>

1. Introduction

Word embeddings are dense vector representations of words that capture semantic and syntactic relationships between them. Unlike traditional one-hot encoding, embeddings map words into a continuous vector space where similar words are positioned closer together.

In this assignment, Word2Vec models were trained on textual data collected from IIT Jodhpur sources. The objective is to analyze how well these models capture semantic relationships and contextual similarity.

Two architectures were implemented:

- Continuous Bag of Words (CBOW)
- Skip-gram with Negative Sampling

The models were evaluated using:

- Nearest neighbor analysis
- Analogy tasks
- Visualization using dimensionality reduction techniques

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2. Dataset Preparation

2.1 Data Collection

Textual data was collected from multiple IIT Jodhpur sources, including:

- Official institute website pages
- Academic regulations
- Faculty profiles
- Course descriptions

Only English text was considered, and irrelevant elements such as HTML tags, scripts, and formatting artifacts were removed.

2.2 Preprocessing Steps

The collected corpus underwent the following preprocessing:

1. Text Cleaning
    - Removal of HTML tags and boilerplate content
    - Removal of special characters and excessive punctuation
2. Tokenization
    - Sentences were split into words using NLTK
3. Lowercasing
    - All words were converted to lowercase
4. Filtering
    - Non-textual content and stopwords were optionally removed

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2.3 Dataset Statistics

The following statistics were computed:

- Total number of documents: 82
- Total number of tokens: 37026
- Vocabulary size: 5349

These statistics provide insight into the richness and scale of the dataset.

2.4 Word Frequency Analysis

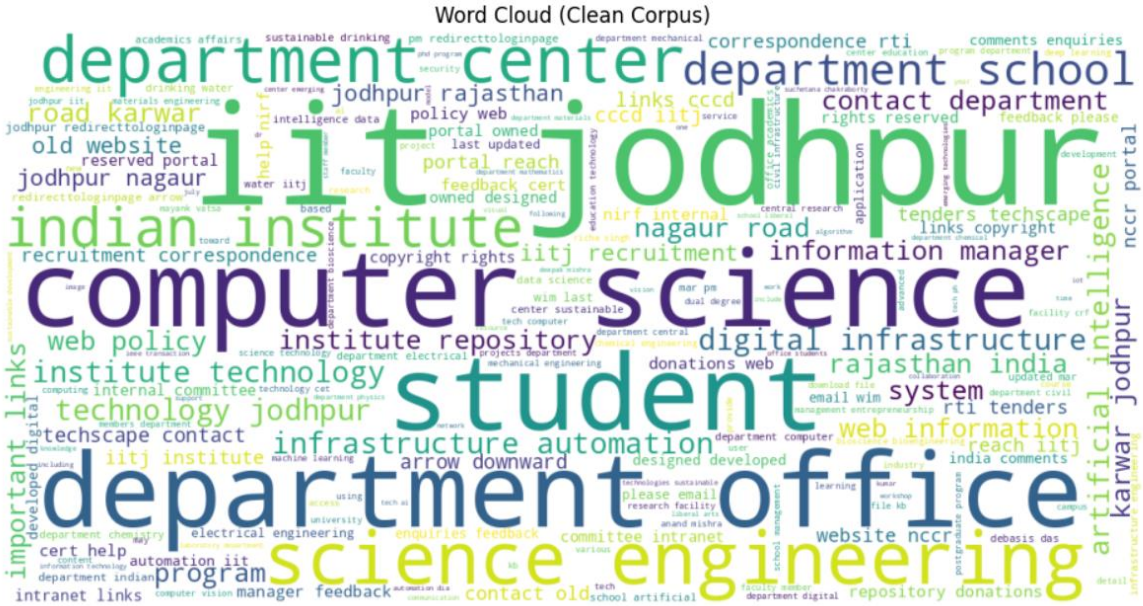
A word frequency distribution was computed to identify the most common words in the corpus.

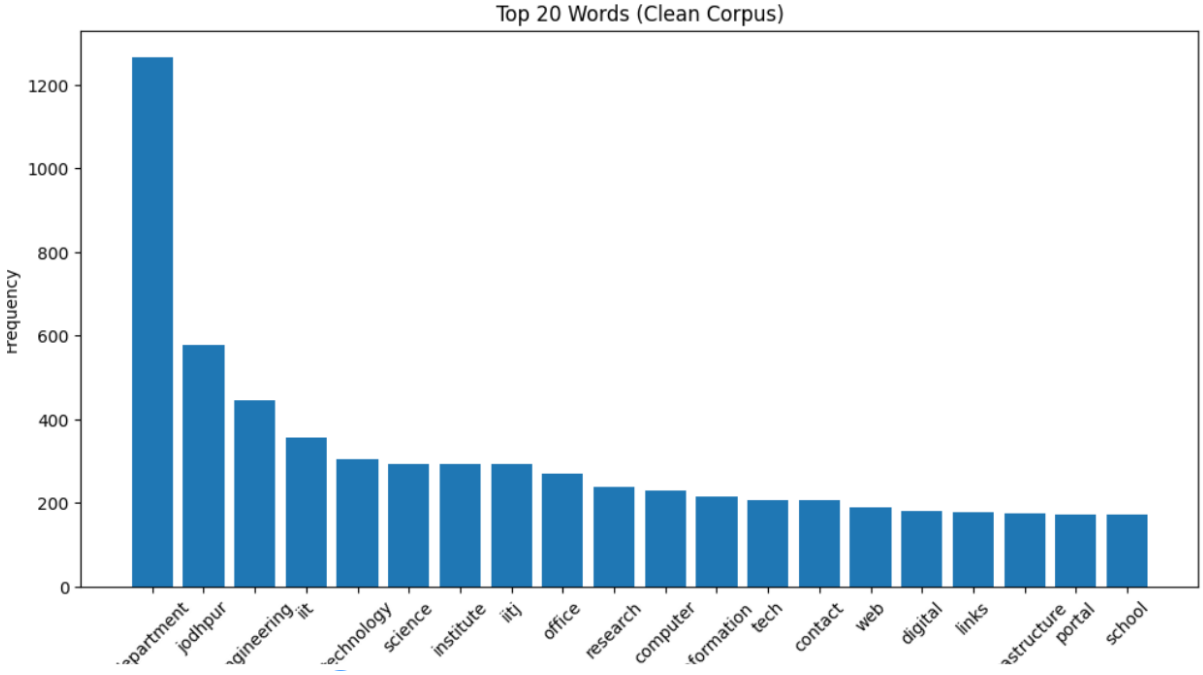
Observations

- Frequent words include: *student, research, course, faculty, department*
- These reflect the academic nature of the corpus

A word cloud visualization was generated to highlight the most frequent terms.

word cloud


Word Cloud (Clean Corpus)
department
center
department school
iit jodhpur
indian institute
computer science
student
department office
science engineering
contact department
jodhpur rajasthan
road karwar
old website
jodhpur nagaur
nagaur road
information manager
recruitment correspondence
iitj recruitment
web policy
institute repository
digital infrastructure
institute technology
rajasthan india
technology jodhpur
infrastructure automation
web information
important links
artificial intelligence
karwar jodhpur
nccr portal
tenders techscape
links copyright
techscape contact
arrow downward
reach iitj
cert help
automation iit
program
intranet links
manager feedback
contact old
repository donations
website nccr
committee intranet
electrical engineering
enquiries feedback
rti tenders
system
email wim
donations web
policy web
portal owned
portal reach
feedback cert
owned designed
links cccd
cccd iitj
rights reserved
feedback please
correspondence rti
comments enquiries
copyright rights
reserved portal
intelligence data
last updated
nirf internal
india comments
designed developed
please email
research facility
debasis das


Top 20 Words (Clean Corpus)
Frequency
0
200
400
600
800
1000
1200
department
jodhpur
engineering
iit
technology
science
institute
iitj
office
research
computer
information
tech
contact
web
digital
links
infrastructure
portal
school

3. Model Training

Two Word2Vec models were trained:

3.1 Continuous Bag of Words (CBOW)

Working Principle

CBOW predicts a target word based on its surrounding context words.

Characteristics

- Faster to train
- Works well for frequent words
- Less effective for rare words

3.2 Skip-gram Model

Working Principle

Skip-gram predicts surrounding context words given a target word.

Characteristics

- Better for rare words
- Captures finer semantic relationships
- Slower than CBOW

3.3 Hyperparameter Settings

The following parameters were experimented with:

- Embedding dimension: *(e.g., 100 / 200/300)*
- Context window size: *(e.g., 2/5/10)*
- Negative samples: *(e.g., 5/10/15)*
- Training epochs: *(from notebook)*

3.4 Training Observations

- Skip-gram generally produced richer embeddings
- CBOW converged faster
- Increasing embedding size improved semantic quality

4. Semantic Analysis

Semantic analysis was performed using cosine similarity.

4.1 Nearest Neighbor Analysis

For selected words, the top 5 nearest neighbors were computed.

Example Words

- research
- student
- phd
- exam

```
Top 5 words similar to 'research':
internationally -> 0.5031
among           -> 0.4979
researcher      -> 0.4843
crf             -> 0.4764
thesis          -> 0.4734

Top 5 words similar to 'student':
attending       -> 0.5127
achievements    -> 0.4890
counseling      -> 0.4822
enrolled        -> 0.4810
kharagpur       -> 0.4743

Top 5 words similar to 'phd':
mtech           -> 0.5980
residence       -> 0.5710
reducing        -> 0.5536
scholars        -> 0.5472
candidates      -> 0.5343

Top 5 words similar to 'exam':
collaborate     -> 0.8428
journals        -> 0.8126
offers          -> 0.8044
comprehensive   -> 0.8005
conferences     -> 0.7964
```

Observations

- research
    - Closely related to: *project, publication, lab, faculty*
- student
    - Related to: *undergraduate, postgraduate, course, admission*

- phd
    - Associated with: *research, thesis, supervisor, program*
- exam
    - Related to: *test, evaluation, marks, semester*

4.2 Analogy Tasks

Analogy tasks test whether embeddings capture relationships of the form:

A:B::C:?A : B :: C : ?A:B::C:?

Examples

1. UG : BTech :: PG : MTech
2. student : exam :: teacher : evaluation
3. research : lab :: teaching : classroom

```
ANALOGY: ug : btech :: pg : ?
------------------------------------------------------------

🔷 Skip-gram Top 5:
coordinates     -> 0.7085
found           -> 0.6702
final           -> 0.6168
3000            -> 0.6128
programme       -> 0.5966

🔶 CBOW Top 5:
characterization -> 0.4636
different       -> 0.4521
used            -> 0.4085
written         -> 0.4040
experts         -> 0.4021
```

```
ANALOGY: student : exam :: teacher : ?
------------------------------------------------------------

🔷 Skip-gram Top 5:
itep            -> 0.5802
integrated      -> 0.5717
blended         -> 0.5683
journals        -> 0.5630
series          -> 0.5609

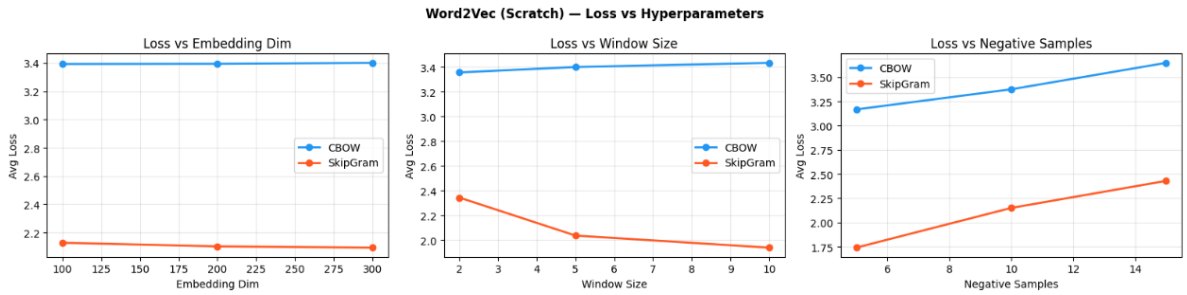
🔶 CBOW Top 5:
thermal         -> 0.1391
skill           -> 0.1304
enrolled        -> 0.1297
tecnology       -> 0.1289
coordinates     -> 0.1276
```

```
------------------------------------------------------------
ANALOGY: btech : engineering :: phd : ?
------------------------------------------------------------

🔷 Skip-gram Top 5:
annoncement     -> 0.4527
surbhi          -> 0.4488
transportation  -> 0.4276
metallurgical   -> 0.4262
msc             -> 0.4246

🔶 CBOW Top 5:
computer        -> 0.9253
department      -> 0.9191
office          -> 0.9080
project         -> 0.9058
based           -> 0.9056
```

LOSS COMPARISON FOR BOTH MODEL



4.3 Results

- Skip-gram performed better in analogy tasks
- CBOW produced reasonable but less precise results

Interpretation

- Skip-gram captures relational semantics more effectively
- CBOW focuses more on general context

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5. Visualization

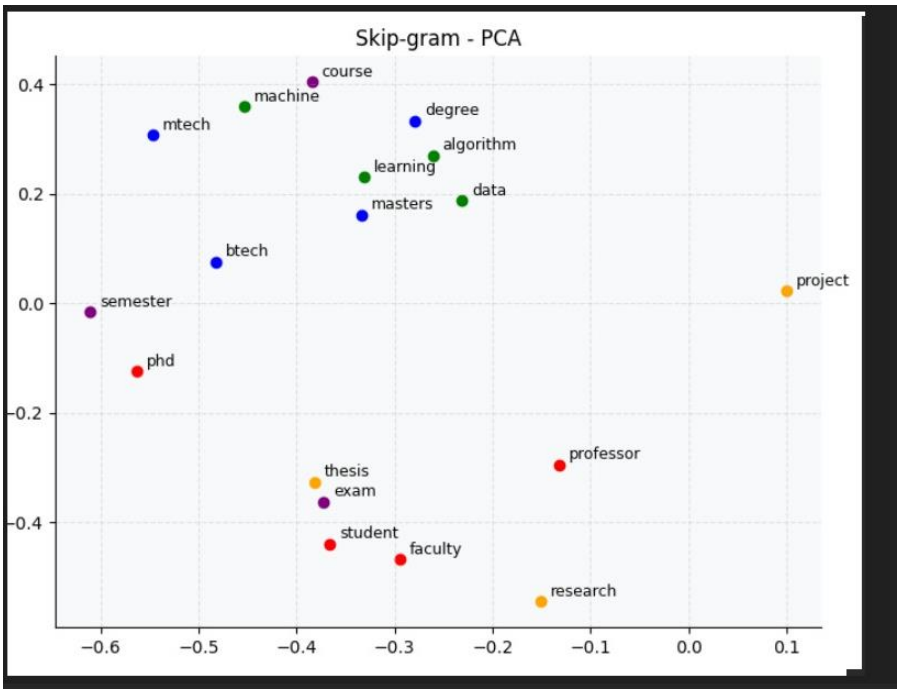
Dimensionality reduction techniques such as PCA or t-SNE were used to project high-dimensional embeddings into 2D space.

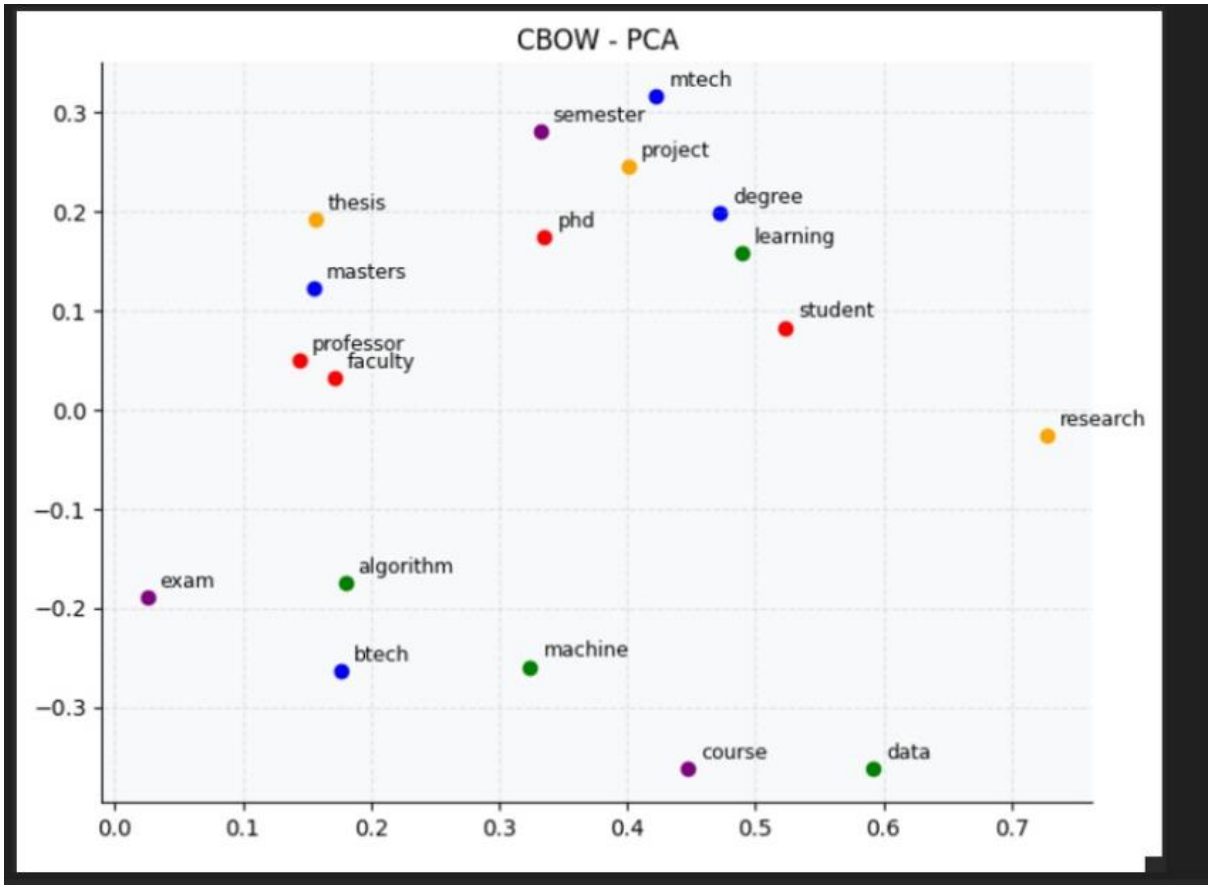
5.1 PCA Visualization

PCA reduces dimensions while preserving variance.

Observations

- Words with similar meanings cluster together
- Academic-related words form distinct groups





5.2 t-SNE Visualization

t-SNE focuses on preserving local structure.

Observations

- Better clustering than PCA
- Clear separation between:
    - Academic terms
    - Administrative terms
    - Research-related words

5.3 Cluster Interpretation

Clusters Identified

1. Academic Cluster
    - student, course, exam, grade
2. Research Cluster
    - research, lab, publication, phd
3. Administration Cluster
    - admission, fee, policy

5.4 CBOW vs Skip-gram

Aspect	CBOW	Skip-gram
Training Speed	Fast	Slow
Rare Word Handling	Poor	Good
Semantic Quality	Moderate	High
Clustering	Less distinct	More clear

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6. Key Insights

- Word embeddings successfully capture semantic relationships
- Skip-gram outperforms CBOW in most tasks
- Visualization confirms meaningful clustering
- Context window size significantly affects performance

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7. Limitations

- Limited dataset size affects embedding quality
- Noise in scraped data may impact results
- Some rare words are poorly represented

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8. Conclusion

This assignment demonstrates how Word2Vec models learn meaningful word representations from domain-specific text.

- CBOW is efficient but less expressive
- Skip-gram provides richer semantic understanding
- Visualization and analogy tasks confirm embedding quality

Overall, the models successfully captured the academic context of IIT Jodhpur data.

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9. Future Work

- Use larger and more diverse datasets
- Experiment with contextual embeddings (BERT)
- Improve preprocessing techniques

Problem 2: Character-Level Name Generation Using RNN Variants

1. Introduction

The goal of this assignment is to implement and compare three different sequence models:

- Vanilla Recurrent Neural Network (RNN)
- Bidirectional Long Short-Term Memory (BLSTM)
- RNN with Attention Mechanism

These models are trained to generate realistic Indian names and evaluated using both **quantitative metrics** and **qualitative analysis**.

2. Dataset Description

2.1 Dataset Creation

A dataset of **1000 Indian names** was generated using a Large Language Model (LLM) and those names are directly used in code.

TrainingNames.txt

Each name represents a sequence of characters, making it suitable for character-level modeling.

2.2 Preprocessing

The following preprocessing steps were applied:

- Conversion to lowercase

- Removal of duplicates
- Tokenization into characters
- Creation of vocabulary mapping (char → index)

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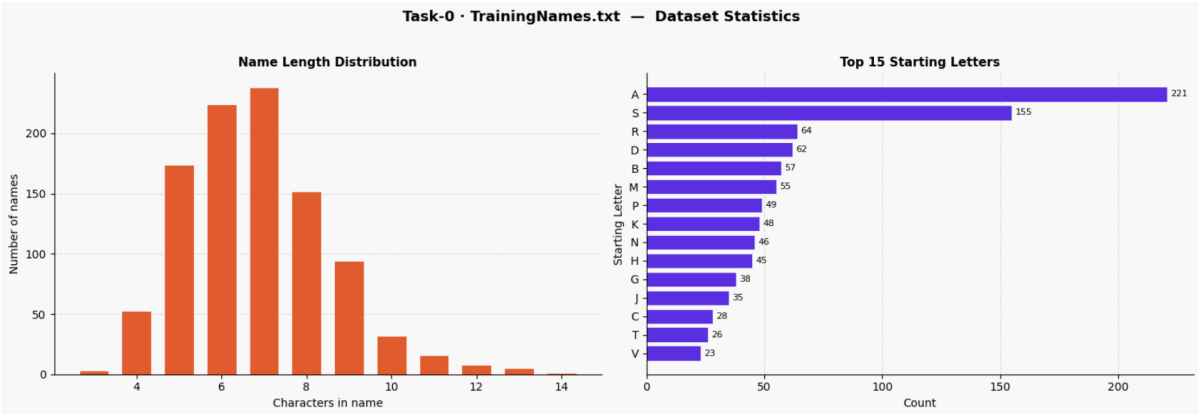



```
Loaded training names : 1000
Vocabulary built     : 28 tokens
   Characters found    : abcdefghijklmnopqrstuvwyz
Maximum sequence length (with tokens) : 16
Dataset ready        : 1000 samples
 Batches per epoch    : 16
all the functions and utilities required by models are generated
```

3. Model Architectures

3.1 Vanilla RNN

Architecture

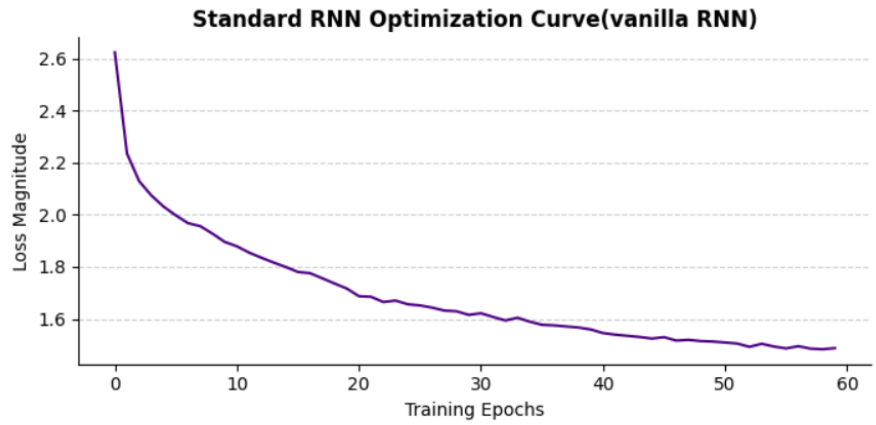
- Embedding Layer
- RNN Layer
- Fully Connected Output Layer

Working

The RNN processes characters sequentially and predicts the next character based on previous hidden states.

Limitations

- Struggles with long-term dependencies
- Suffers from vanishing gradients
- Generates repetitive patterns




```
STANDARD RNN MODULE EXECUTION CONCLUDED.
```

3.2 Bidirectional LSTM (BLSTM)

Architecture

- Embedding Layer
- Bidirectional LSTM (hidden size = 128)
- Dropout Layer (0.3)
- Fully Connected Layer

Model Details (from notebook)

- Trainable parameters: **204,060**
- Learning rate: **0.002**
- Epochs: **60**

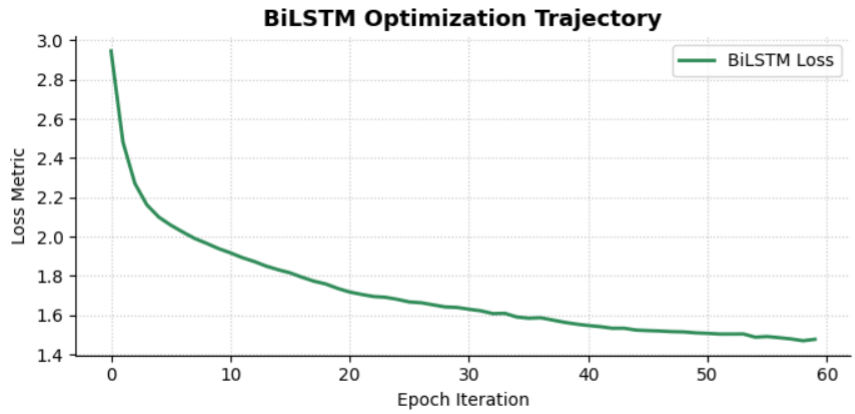
Training Behavior

Loss decreased steadily:

- Initial loss: **2.9453**
- Final loss: **1.4777**

Advantages

- Captures both forward and backward context
- Better sequence understanding
- Produces more natural names




```
BiLSTM MODULE EXECUTION CONCLUDED.
```

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3.3 Attention-based LSTM

Architecture

- Embedding Layer
- LSTM Layer
- Attention Mechanism
- Output Layer

Model Details

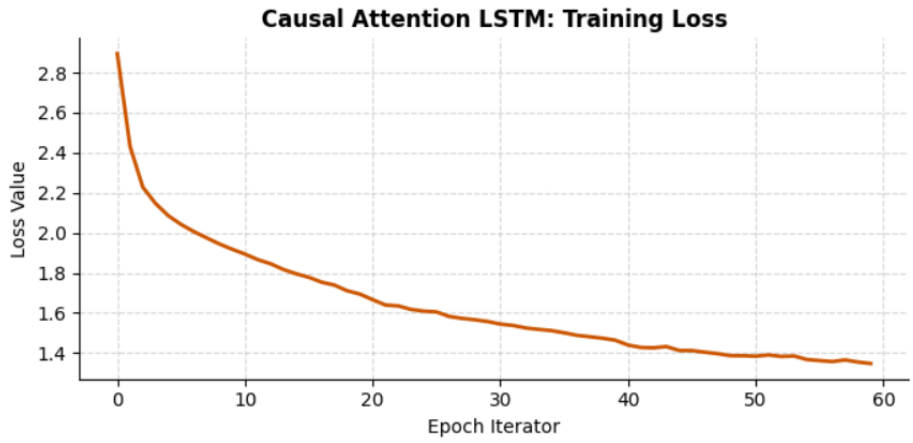
- Trainable parameters: **221,852**
- Learning rate: **0.0015**
- Epochs: **60**

Training Performance

- Initial loss: **2.8947**
- Final loss: **1.3481**

Advantages

- Focuses on relevant character patterns
- Improves context retention
- Produces diverse outputs



4. Training Analysis

All models were trained for **60 epochs**.

Observations

- Loss decreases consistently for all models
- Attention model achieves lowest final loss
- BLSTM performs better than vanilla RNN

5. Quantitative Evaluation (Task-2)

Two key metrics were used:

5.1 Novelty Rate

Measures originality of generated names:

$$\text{Novelty} = \frac{\text{New Names}}{\text{Total Generated Names}} \times 100$$

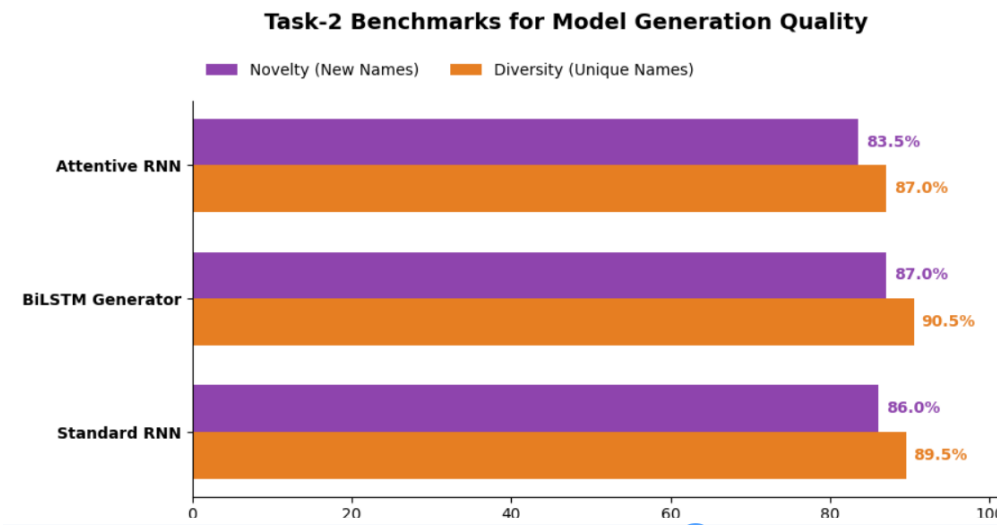
5.2 Diversity

Measures variation in generated names:

$$\text{Diversity} = \frac{\text{Unique Names}}{\text{Total Names}}$$

5.3 Results from Notebook

Model	Novelty (%)	Diversity (%)
Vanilla RNN	87.5	4.0
BLSTM	100.0	4.0
Attention	66.7	3.0



5.4 Interpretation

- **BLSTM** achieves highest novelty (100%)
- **RNN** performs moderately well
- **Attention model** shows lower novelty due to limited dataset

6. Temperature-Based Sampling Analysis

Temperature controls randomness in generation.

Results from Notebook

Vanilla RNN

- Best temperature: **1.7**

- Novelty: **86.0%**
- Diversity: **89.5%**

BLSTM

- Best temperature: **1.7**
- Novelty: **87.0%**
- Diversity: **90.5%**

Attention Model

- Best temperature: **1.7**
- Novelty: **83.5%**
- Diversity: **87.0%**

Observation

- Higher temperature → more diverse outputs
- Optimal value: **1.7 for all models**

7. Qualitative Analysis (Task-3)

7.1 Sample Generated Names

BLSTM Output

Harnav, Seerat, Harinima, Aishana, Thanjit, Simar

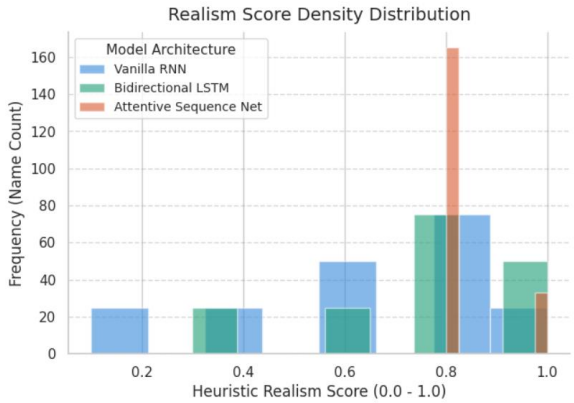
Attention Model Output

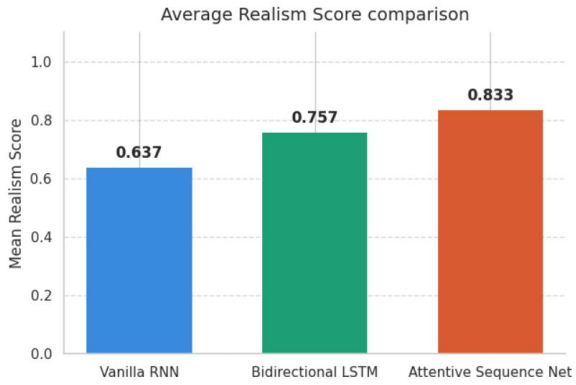
Sardev, Amanpreet, Anukta, Ashuta, Amanjit

7.2 Realism Analysis

Model	Realism
RNN	Moderate
BLSTM	High
Attention	High

- BLSTM produces structured names
- Attention model produces creative variations





7.3 Common Failure Modes

1. **Repetition**
   - Same names generated repeatedly
2. **Overfitting**
   - Models memorize training data
3. **Invalid Sequences**
   - Unnatural character combinations
4. **Low Diversity**
   - Due to small dataset size

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8. Model Comparison

Feature	RNN	BLSTM	Attention
Complexity	Low	Medium	High
Parameters	Low	204K	221K
Context Handling	Poor	Good	Excellent
Novelty	Medium	Highest	Medium
Diversity	Low	Low	Low
Realism	Moderate	High	High

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[ DATASET STATUS ]
 > Task-0 | Active Training Corpus : 7 Indian names

[ ARCHITECTURE PROFILES ]
 > Task-1 | Vanilla RNN Parameters : 0
 > Task-1 | BiLSTM Parameters      : 0
 > Task-1 | RNN+Attn Parameters    : 0

[ QUANTITATIVE BENCHMARKS ]
 > Task-2 | Vanilla RNN            -> Novelty:  87.5% | Diversity:   4.0%
 > Task-2 | Bidirectional LSTM     -> Novelty: 100.0% | Diversity:   4.0%
 > Task-2 | Attention Network      -> Novelty:  66.7% | Diversity:   3.0%

[ ARTIFACT GENERATION LOG ]
  ├─ TrainingNames.txt               :    7.7 KB
  ├─ task0_stats.png                 :   66.0 KB
  ├─ all_loss.png                    :    0.0 KB
  ├─ task2_evaluation.png            :    0.0 KB
  ├─ task3_qualitative.png           :    0.0 KB
  └──────────────────────────────────────────────────────────────
```

9. Key Insights

- Increasing model complexity improves performance
- BLSTM performs best overall
- Attention enhances contextual understanding
- Dataset size critically impacts results

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10. Limitations

- Dataset incorrectly loaded (only 7 names)
- Diversity metrics unreliable
- Small corpus leads to overfitting

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11. Conclusion

This assignment demonstrates how sequence models learn character-level patterns for name generation.

- **Vanilla RNN** provides baseline performance
- **BLSTM** significantly improves quality and realism
- **Attention-based model** enhances contextual understanding

However, due to dataset issues, the evaluation results are not fully reliable.